

Cover Letter

Candidate: Ahmet Halit Ünsal

Contact: ahmethalitunsal@gmail.com | +90 (552) 232 81 44 | Istanbul, Turkey

Position: Head of Powertrain & ECU Embedded Software Architecture

Target Organization: TUSAŞ Motor Sanayii (TEI) (Eskişehir / Ankara, Turkey)

Job URL: <https://www.tei.com.tr/kariyer/ecu-software-lead-401>

Dear Hiring Team & Leadership at TUSAŞ Motor Sanayii (TEI),

I am writing to express my strong interest in the Head of Powertrain & ECU Embedded Software Architecture position at TUSAŞ Motor Sanayii (TEI).

With over 15 years of professional engineering experience - including 8+ years leading and directing multi-disciplinary engineering organizations of up to 30 engineers across 4 team leads - my background directly aligns with TUSAŞ Motor Sanayii (TEI)'s requirements for high-reliability embedded software architecture, Model-Based Design (MBD), and mission-critical powertrain controls.

Throughout my leadership tenure across automotive and defense electronics programs (NISO Technology, ECEMTAG, TÜBİTAK, TÜMOSAN), I have directed full product lifecycles from initial concept and dynamic modeling to dyno characterization and mass production. I maintain strict enforcement of ISO 26262 ASIL D functional safety, ASPICE Level 2/3 processes, UN R155/R156 cybersecurity (CSMS), and AUTOSAR layered architectures across FreeRTOS and bare-metal platforms.

On the technical front, I bring hands-on mastery in MATLAB/Simulink/Stateflow physical modeling, PMSM/IPMSM Field-Oriented Control (FOC, MTPA, flux-weakening), and complex EV powertrain controls (VCU, MCU, Traction Inverters, OBC, BMS). Complementing this is extensive experience establishing automated Hardware-in-the-Loop (HIL) simulation testbenches (dSpace MABX, Lauterbach Trace32) and Git-based CI/CD pipelines that dramatically compress verification cycles and eliminate integration bottlenecks.

Beyond embedded systems, my end-to-end technical leadership is demonstrated in platforms like EduTrace (edutrace.net), highlighting my ability to scale complex architectures and bridge software, hardware, and product teams. I would welcome the opportunity to discuss how my technical ownership, executive leadership, and delivery focus can accelerate the engineering milestones at TUSAŞ Motor Sanayii (TEI).

Sincerely,

Ahmet Halit Ünsal

<https://www.ahmethalitunsal.com> | <https://www.linkedin.com/in/ahmet-halit-unsal/> | <https://github.com/aunsal89>